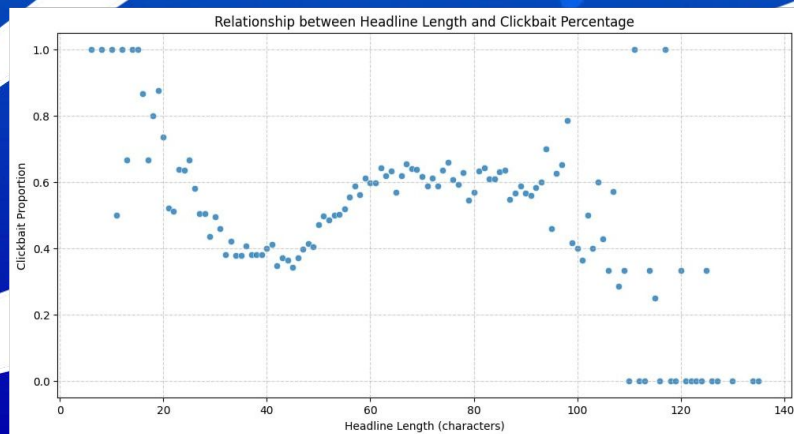
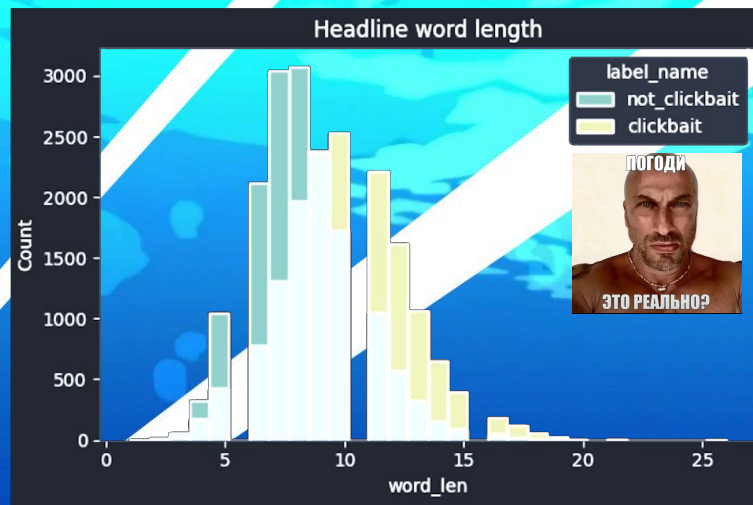




Clickbait

Задача

train 90
val 5
test 5



Классические методы

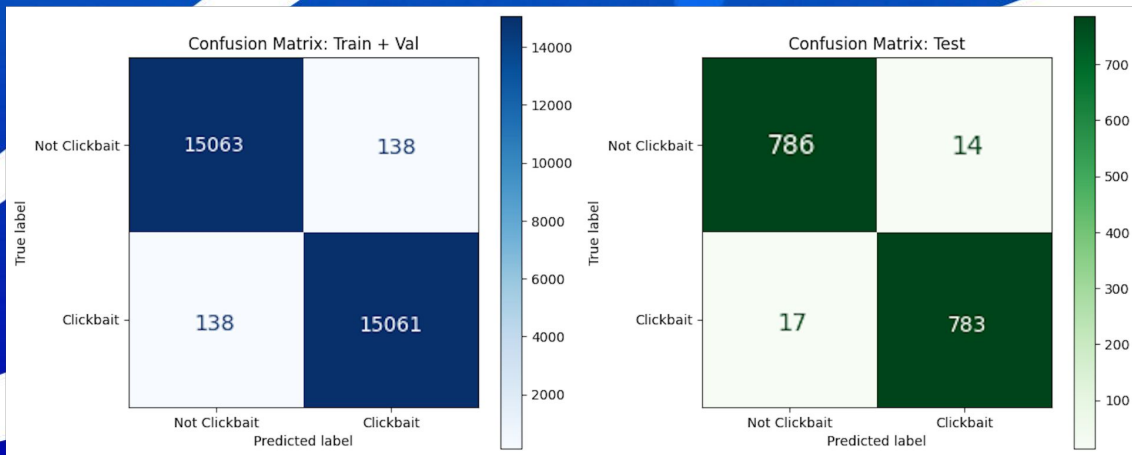
Ручной feature engineering:

- ▶ TF-IDF (Word Level, Char Level)
- ▶ TextBlob и Vader
- ▲ Индекс удобочитаемости
- ◀ Флеша, Флеша-Кинкейда
- ◀ Эвристики

(Доля символов/слов
в верхнем регистре,
количество цифр и т.д.)

Модель:
LogisticRegression

	Train + Val	Test
F1-Score	0.9909	0.9806



Нейросети RNN

method	test_accuracy	test_f1	notes
RNN/GRU benchmark	0.9612	0.961000	Teammate benchmark, split 28800/1600/1600
LLM zero-shot	0.5000	0.074074	Measured on 50 examples
LLM + QLoRA	0.9800	0.980198	Measured on full test split: 200 examples

TEST ACCURACY

96.12%

TEST F1

96.10%

true \ pred
not_clickbait

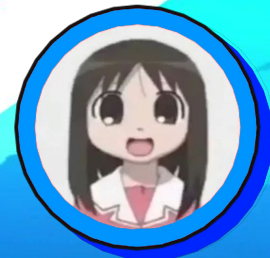
775

25

clickbait

37

763

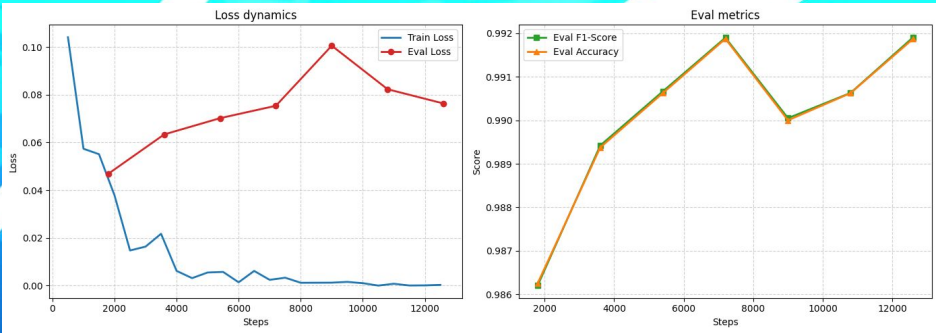


distilbert/distilbert-base-uncased

Плюсы ✨:

1. Универсально
2. Эффективно
3. Быстро
4. XAI

Metric	Test Score
Accuracy	0.988125
Precision	0.991195
Recall	0.985000
F1-Score	0.988088



Legend: ■ Negative ■ Neutral ■ Positive

True Label	Predicted Label	Attribution Label	Attribution Score
Clickbait	Clickbait (1.00)	Clickbait	2.32

Word Importance

[CLS] michael bay has made a real tragedy into a dumb action movie [SEP]

Нейросети LoRA

Использовать? а ты мне ответишь Qwen/Qwen2.5-1.5B-Instruct

Ок.



Какую модель использовать?

Qwen/Qwen2.5-1.5B-Instruct



Нейросети LoRA

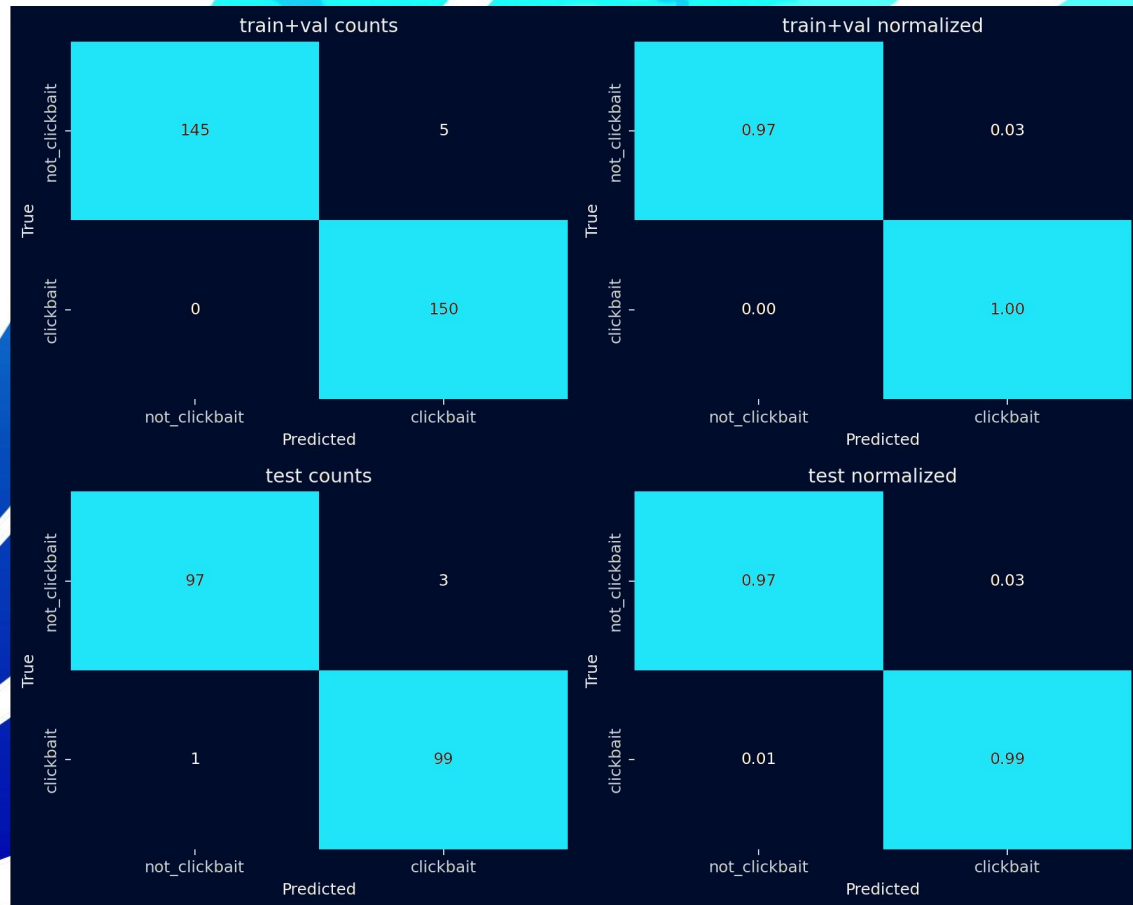
```
MAX_LENGTH = 160
```

```
inputs = tokenizer(  
    prompts,  
    return_tensors="pt",  
    padding=True,  
    truncation=True,  
    max_length=MAX_LENGTH,  
)
```

	headline	label	answer
0	Iran says its nuclear program is unchanged	0	A
1	It's A Turtleneck Battle: Drake Vs. Nick Jonas	1	B
2	This Photographer Tied A Model Up In Shark-Inf...	1	B
3	13 killed in U.S. air strike in Pakistan	0	A
4	Indian Markets hits record close in Diwali trade	0	A

Нейросети LoRA

split	metric	value
train+val	accuracy	0.983333
train+val	precision	0.967742
train+val	recall	1.000000
train+val	f1	0.983607
test	accuracy	0.980000
test	precision	0.970588
test	recall	0.990000
test	f1	0.980198



5070.4s 110 Validation metrics: {'eval_loss': 0.0034949074033647776, 'eval_ab_accuracy': 0.998125, 'eval_ab_precision': 0.9975031210986267, 'eval_ab_recall': 0.99875, 'eval_ab_f1': 0.9981261711430356, 'eval_ab_invalid_labels': 0.0}

Нейросети

Prompt engineering

Model	limited summary				multioutput structured summary				structured summary				few shot system summary			
	Accuracy	F1-Score	Precision	Recall	Accuracy	F1-Score	Precision	Recall	Accuracy	F1-Score	Precision	Recall	Accuracy	F1-Score	Precision	Recall
gemma_4_e4b	0.60	0.67	0.50	1.00	0.60	0.50	0.50	0.50	0.70	0.73	0.57	1.00	0.90	0.89	0.80	1.00
qwen_3_8b	0.70	0.57	0.67	0.50	0.70	0.67	0.60	0.75	0.90	0.89	0.80	1.00	0.80	0.75	0.75	0.75
gemma_4_e2b	0.50	0.58	0.44	0.88	0.40	0.33	0.30	0.38	0.80	0.80	0.67	1.00	0.90	0.88	0.88	0.88
gemma_3_4b	0.65	0.36	0.67	0.25	0.40	0.45	0.36	0.62	0.70	0.73	0.57	1.00	0.60	0.67	0.50	1.00
deepseek_r1_1_5b	0.62	0.51	0.71	0.40	0.42	0.41	0.42	0.40	0.54	0.55	0.54	0.56	0.34	0.35	0.35	0.36
gemma_2_9b	0.70	0.40	1.00	0.25	0.70	0.67	0.60	0.75	0.80	0.75	0.75	0.75	0.80	0.75	0.75	0.75
stablelm2_1_6b	0.50	0.07	0.50	0.04	0.46	0.56	0.47	0.68	0.50	0.67	0.50	1.00	0.58	0.68	0.55	0.88
smollm3_3b	0.75	0.67	0.71	0.62	0.50	0.50	0.42	0.62	0.80	0.80	0.67	1.00	0.60	0.67	0.50	1.00
granite_3_0_8b	0.40	0.17	0.27	0.12	0.42	0.38	0.41	0.36	0.74	0.79	0.67	0.96	0.90	0.91	0.83	1.00
yi_1_5_9b	0.70	0.73	0.57	1.00	0.40	0.40	0.33	0.50	0.60	0.67	0.50	1.00	0.80	0.80	0.67	1.00
stablelm2_12b	0.40	0.57	0.40	1.00	0.90	0.89	0.80	1.00	0.80	0.75	0.75	0.75	0.80	0.80	0.67	1.00
mistral_7b	0.30	0.22	0.20	0.25	0.40	0.50	0.38	0.75	0.70	0.57	0.67	0.50	0.90	0.89	0.80	1.00

Нейросети

Prompt engineering

Model	limited summary				multioutput structured summary				structured summary				zero shot summary			
	Accuracy	F1-Score	Precision	Recall	Accuracy	F1-Score	Precision	Recall	Accuracy	F1-Score	Precision	Recall	Accuracy	F1-Score	Precision	Recall
gemma_4_e4b	0.80	0.75	0.75	0.75	0.60	0.50	0.50	0.50	0.80	0.75	0.75	0.75	0.40	0.50	0.38	0.75
qwen_3_8b	0.80	0.67	1.00	0.50	0.50	0.44	0.40	0.50	0.50	0.55	0.43	0.75	0.70	0.57	0.67	0.50
gemma_4_e2b	0.90	0.88	0.88	0.88	0.25	0.21	0.18	0.25	0.65	0.67	0.54	0.88	0.85	0.82	0.78	0.88
gemma_3_4b	0.40	0.57	0.40	1.00	0.50	0.44	0.40	0.50	0.60	0.64	0.50	0.88	0.50	0.58	0.44	0.88
deepseek_r1_1_5b	0.52	0.54	0.52	0.56	0.34	0.38	0.36	0.40	0.58	0.66	0.56	0.80	0.66	0.62	0.70	0.56
gemma_2_9b	0.30	0.46	0.33	0.75	0.50	0.44	0.40	0.50	0.80	0.75	0.75	0.75	0.80	0.75	0.75	0.75
stablelm2_1_6b	0.44	0.52	0.45	0.60	0.46	0.40	0.45	0.36	0.54	0.63	0.53	0.80	0.60	0.62	0.59	0.64
smollm3_3b	0.65	0.59	0.56	0.62	0.45	0.35	0.33	0.38	0.75	0.67	0.71	0.62	0.70	0.50	0.75	0.38
granite_3_0_8b	0.66	0.73	0.61	0.92	0.42	0.43	0.42	0.44	0.70	0.72	0.68	0.76	0.64	0.50	0.82	0.36
yi_1_5_9b	0.40	0.57	0.40	1.00	0.50	0.44	0.40	0.50	0.80	0.80	0.67	1.00	0.60	0.67	0.50	1.00
stablelm2_12b	0.60	0.33	0.50	0.25	0.50	0.44	0.40	0.50	0.80	0.67	1.00	0.50	0.50	0.44	0.40	0.50
mistral_7b	0.80	0.75	0.75	0.75	0.50	0.44	0.40	0.50	0.70	0.73	0.57	1.00	0.80	0.75	0.75	0.75

system prompt summary			
Accuracy	F1-Score	Precision	Recall
0.80	0.75	0.75	0.75
0.70	0.67	0.60	0.75
0.85	0.82	0.78	0.88
0.50	0.62	0.44	1.00
0.54	0.57	0.54	0.60
0.80	0.75	0.75	0.75
0.46	0.47	0.46	0.48
0.65	0.46	0.60	0.38
0.70	0.63	0.81	0.52
0.80	0.80	0.67	1.00
0.60	0.60	0.50	0.75
0.50	0.62	0.44	1.00



